

AORANE

Multi-Agent System Design

How AI agents using platforms like n8n can automate operations across Sales, Support, Marketing, Finance & Engineering.

10

AI Agents

~ ¹9,500

Monthly Cost

¹2.2L

Salary Saved / mo

23x

ROI

WHAT THIS DOCUMENT COVERS

- Full multi-agent architecture diagram with data flow
- 10 individual AI agents — purpose, workflow steps, tools used
- Platform comparison: n8n vs Make vs Zapier vs custom build
- Detailed monthly cost breakdown (INR + USD)
- Human effort replaced and ROI calculation
- Implementation roadmap — Phase 1 vs Phase 2

What Are AI Agents & Why AORANE Needs Them

An AI Agent is an autonomous software program that perceives a trigger (event, webhook, schedule), reasons about it using an AI model, executes a sequence of actions (API calls, messages, data writes), and completes a task — without human intervention. Platforms like n8n provide a visual canvas to connect these steps.

The Problem Without Agents

- ' Every new user signup requires manual welcome emails and onboarding messages
- ' Failed payments go unrecovered for days without active human follow-up
- ' B2B leads from enquiry forms wait hours before a sales rep responds
- ' Health content for social media requires a dedicated writer every day
- ' Blood emergency alerts depend on a human seeing the notification
- ' Monthly health reports for business clients need manual compilation

The Solution With AI Agents

- ' Signup triggers an automated 7-day onboarding sequence instantly
- ' Razorpay webhook fires the payment recovery agent within 60 seconds
- ' Lead enquiry !' AI-qualified !' BDM notified on Slack in under 2 minutes
- ' Daily content agent generates + publishes health posts at 8am automatically
- ' Blood emergency agent blasts SMS + push to nearby donors in under 30 seconds
- ' Analytics agent generates and emails business reports every Monday 7am

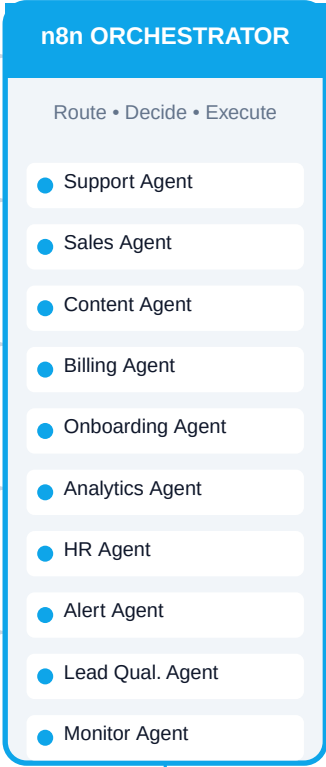
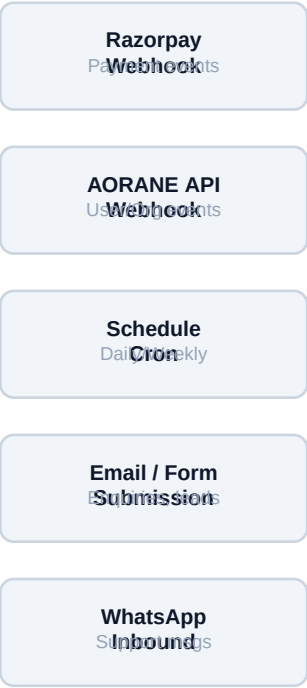
Platform Comparison

Criteria	n8n (Self-host)	Make.com	Zapier	Custom Build
Monthly Cost	£1,200 (VPS)	£3,500+	£8,000+	£15,000+
AI Model Support	' Any (OpenAI, Gemini, NVIDIA)	' Limited	' Limited	' Full
Custom Logic	' Full JS/Python code	%³ Limited	' Basic	' Full
Workflow Executions	Unlimited	10K/mo	750/mo (starter)	Unlimited
Data Privacy	' Self-hosted, DPDPA OK	%³ Cloud only	%³ Cloud only	' Full
Setup Time	1–2 days	1 day	< 1 day	4–8 weeks
AORANE Fit	+P BEST FIT	%³ OK	' Too expensive	%³ Later stage

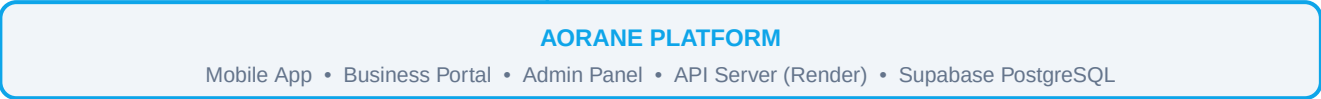
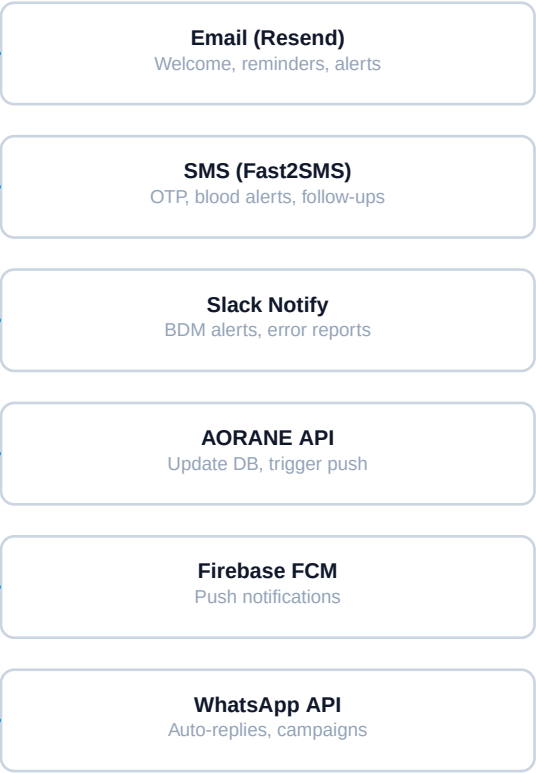
Multi-Agent Architecture Diagram

Data flow: Triggers !' n8n Orchestrator !' AI Agents !' Actions !' AORANE Platform

TRIGGER SOURCES

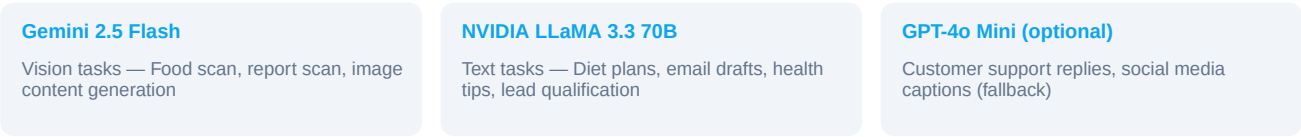


ACTIONS & OUTPUTS



LEGEND: — Primary data flow (from n8n) — Trigger !' Orchestrator feed

AI MODELS POWERING THE AGENTS:



The 10 AI Agents — Detailed Design

1 Customer Support Agent

TRIGGER: New WhatsApp/email message or support ticket created

WORKFLOW: 1. Classify intent (refund / technical / general) !' 2. Search AORANE knowledge base !' 3. Generate reply using Gemini !' 4. Send reply; if unresolved !' escalate to human

TOOLS: [AORANE API](#) • [Gemini 2.5 Flash](#) • [Resend Email](#) • [WhatsApp API](#)

REPLACES: 1 Support Executive (₹18,000– ₹30,000/month)

2 B2B Sales Outreach Agent

TRIGGER: New company added to CRM or weekly Monday 9am schedule

WORKFLOW: 1. Fetch prospect data from HubSpot/sheet !' 2. Research company via web search !' 3. Draft personalized pitch email using LLaMA 70B !' 4. Send email !' log in CRM !' set 3-day follow-up reminder

TOOLS: [HubSpot / Google Sheets](#) • [NVIDIA LLaMA 70B](#) • [Resend / Gmail SMTP](#) • [Slack \(BDM notify\)](#)

REPLACES: 1 Inside Sales Executive cold outreach effort (₹30,000– ₹55,000/month)

3 Content & Social Media Agent

TRIGGER: Daily at 8:00am IST (scheduled cron)

WORKFLOW: 1. Pick health topic for the day from content calendar !' 2. Generate post copy (Hindi+English) via LLaMA 70B !' 3. Create caption + hashtags !' 4. Publish to Instagram + LinkedIn + save draft for Twitter/X

TOOLS: [NVIDIA LLaMA 70B](#) • [Instagram Graph API](#) • [LinkedIn API](#) • [Buffer / Social scheduler](#)

REPLACES: 1 Social Media Manager daily posting effort (₹25,000– ₹42,000/month)

4 Payment Recovery Agent

TRIGGER: Razorpay webhook: payment_failed or subscription expired

WORKFLOW: 1. Identify user from payment ID via AORANE API !' 2. Check failure reason (card/UPI/bank) !' 3. Send personalized recovery email + SMS within 60 seconds !' 4. Retry on Day 1, Day 3, Day 7 !' cancel subscription if all fail

TOOLS: [Razorpay API](#) • [AORANE API](#) • [Resend Email](#) • [Fast2SMS](#)

REPLACES: Manual billing follow-up effort — recovers ~20–35% of failed payments automatically

5 User Onboarding Agent

TRIGGER: AORANE API webhook: new_user_registered or new_org_created

WORKFLOW: 1. Send instant welcome email with app download link !' 2. Day 3: Send health tip + food log tutorial nudge !' 3. Day 7: Send diet plan activation prompt + Pro upgrade offer !' 4. Day 30: Send retention survey + referral program invite

TOOLS: [AORANE API](#) • [Resend Email](#) • [Fast2SMS](#) • [Firebase FCM](#)

REPLACES: 1 Customer Success Executive manual onboarding (₹30,000– ₹50,000/month)

The 10 AI Agents — Continued (Agents 6–10)

6 Analytics & Reporting Agent

TRIGGER: Every Monday 7:00am + 1st of each month

WORKFLOW: 1. Query Supabase for weekly user metrics (DAU, MAU, new signups) !' 2. Generate B2B org health summary report per organization !' 3. Summarize insights using Gemini (natural language) !' 4. Email PDF report to business portal admins + internal Slack

TOOLS: [Supabase API](#) • [AORANE API](#) • [Gemini 2.5 Flash](#) • [Resend Email](#) • [Slack](#)

REPLACES: Manual report compilation by Product Manager or Analyst (£180,000– £1,40,000/month PM time)

7 HR Recruitment Agent

TRIGGER: New job application email received or job post published on LinkedIn

WORKFLOW: 1. Parse CV text using Gemini AI !' 2. Score against job requirements (tech stack match, experience) !' 3. Draft shortlisting email or rejection (politely) !' 4. Schedule interview via Calendly link; notify hiring manager on Slack

TOOLS: [Gemini 2.5 Flash](#) • [Gmail / Resend](#) • [Calendly API](#) • [Slack](#) • [Google Sheets \(ATS\)](#)

REPLACES: 50% of HR Recruiter screening time (£140,000– £170,000/month HR cost)

8 Blood Emergency Alert Agent

TRIGGER: AORANE API webhook: blood_emergency_request_created

WORKFLOW: 1. Extract blood group + location from emergency request !' 2. Query donor database for matching blood group within 50km radius !' 3. Send SMS blast to top 20 donors via Fast2SMS instantly !' 4. Send Firebase FCM push to donor app users within range

TOOLS: [AORANE API](#) • [Fast2SMS](#) • [Firebase FCM](#) • [Google Maps API \(distance calc\)](#)

REPLACES: Manual alert process — reduces alert delivery time from hours to < 30 seconds

9 Lead Qualification Agent

TRIGGER: New enquiry form submission on aorane.com landing page

WORKFLOW: 1. Extract company name, size, city, email from form data !' 2. Score lead: company size > 50 employees = High, < 50 = Medium !' 3. Enrich lead with LinkedIn company data !' 4. Notify BDM on Slack with score + draft intro email ready to send

TOOLS: [AORANE API](#) • [NVIDIA LLaMA 70B](#) • [Slack](#) • [Hunter.io / Apollo.io](#) • [HubSpot CRM](#)

REPLACES: Pre-sales qualification effort — 1 Pre-Sales Specialist (£135,000– £160,000/month)

10 API Monitoring & Incident Agent

TRIGGER: Every 5 minutes (cron) + Render/Vercel error webhook

WORKFLOW: 1. Ping all AORANE API health endpoints (/api/health, /api/plans etc.) !' 2. If response > 3s or status " 200 !' classify severity !' 3. Send PagerDuty / Slack alert to on-call engineer immediately !' 4. Auto-document incident in GitHub Issue for post-mortem

TOOLS: [HTTP Requests](#) • [Render API](#) • [Slack / PagerDuty](#) • [GitHub API](#)

REPLACES: Dedicated DevOps monitoring effort — catches issues before users report them

Full Cost Breakdown & ROI Analysis

Section A — Infrastructure & Platform Costs (Monthly)

Item	Plan / Detail	USD/mo	INR/mo
n8n (Self-hosted) on Hetzner VPS	CX21 — 2vCPU 4GB RAM	\$5	₹420
Hetzner VPS (n8n + monitoring)	Helsinki server, low latency	\$6	₹500
n8n Cloud (alternative, optional)	Starter — 5K executions	\$20	₹1,670
Domain / SSL for n8n instance	Cloudflare, Let's Encrypt	\$1	₹85
Total Infrastructure	(self-hosted option)	\$12	₹1,000

Section B — AI Model API Costs (Monthly)

Item	Plan / Detail	USD/mo	INR/mo
Gemini 2.5 Flash API	~2M input tokens/month (vision tasks)	\$0.15	₹13
NVIDIA LLaMA 3.3 70B	Already available via NVIDIA API KEY	\$0	₹0
OpenAI GPT-4o Mini (optional fallback)	500K tokens/month for support agent	\$0.15	₹13
Gemini Pro (for reports, optional)	500K tokens for detailed summaries	\$0.63	₹53
Total AI API Cost		\$0.93 – \$1.5	₹80 – ₹125

Section C — Third-Party Tool Costs (Monthly)

Item	Plan / Detail	USD/mo	INR/mo
WhatsApp Business API (Interakt/Wati)	1000 conversations/month included	\$15	₹1,250
Hunter.io (lead enrichment)	Starter 50 searches/month	\$0	₹0 (free tier)
HubSpot CRM	Free tier (up to 1M contacts)	\$0	₹0
Calendly (HR interviews)	Basic free tier	\$0	₹0
Fast2SMS (SMS alerts)	~2000 SMS/month for blood + OTP	\$0	₹600
PagerDuty (monitoring alerts)	Free tier (1 user)	\$0	₹0
Slack (team notifications)	Free tier (90-day history)	\$0	₹0
Total Tool Cost		\$15	₹1,850

GRAND TOTAL MONTHLY COST FOR ALL 10 AI AGENTS

₹2,930 – ₹3,375 / month (~\$35 – \$40 USD)

ROI Analysis & Implementation Roadmap

Human Effort Replaced by AI Agents

Agent	Role Replaced	Salary (\$/mo)	Coverage %	Saving (\$/mo)
Support Agent (#1)	Support Executive	\$24,000	70%	\$16,800
Sales Outreach (#2)	Inside Sales SDR	\$42,500	50%	\$21,250
Content Agent (#3)	Social Media Mgr	\$33,500	80%	\$26,800
Payment Recovery (#4)	Billing Follow-up	\$27,500	100%	\$27,500
Onboarding Agent (#5)	Customer Success	\$40,000	90%	\$36,000
Analytics Agent (#6)	PM/Analyst time	\$1,10,000	20%	\$22,000
HR Agent (#7)	HR Recruiter	\$55,000	40%	\$22,000
Blood Alert Agent (#8)	Manual process	\$0 (prev. none)	100%	\$8,000
Lead Qual. Agent (#9)	Pre-Sales Exec	\$47,500	60%	\$28,500
Monitor Agent (#10)	DevOps on-call	\$80,000	30%	\$24,000

TOTAL SAVINGS		\$2,32,850 / month	
Monthly AI Cost	Monthly Saving	Net Monthly Saving	Return on Investment
\$13,375	\$12,32,850	\$12,29,475	68x
All 10 agents running	Salary + effort replaced	Saving minus agent cost	Every \$1 spent = \$68 saved

Implementation Roadmap

Week 1–2	Setup & Quick Wins
- Blood Alert Agent (#8) — highest impact, life-saving - Payment Recovery Agent (#4) — immediate revenue recovery - API Monitor Agent (#10) — prevent downtime	
Week 3–4	User Lifecycle Automation
- Onboarding Agent (#5) — improve activation & retention - Analytics Agent (#6) — automate weekly business reports - Lead Qualification Agent (#9) — faster B2B pipeline	
Month 2	Growth & Sales Automation
- Content Agent (#3) — daily social media publishing - Sales Outreach Agent (#2) — systematic B2B prospecting - Customer Support Agent (#1) — reduce ticket volume	
Month 3	Full Automation
- HR Recruitment Agent (#7) — CV screening + scheduling - Fine-tune all agents based on 2-month performance data - Add new trigger types as business grows	

Setup Time Estimate: 1 developer, part-time (2–3 hours/day) can deploy all 10 agents in 3–4 weeks using n8n's visual canvas. No new backend code needed — agents call existing AORANE APIs.

